Equations of Straight Lines (y = mx + c)

Exam code: 0607

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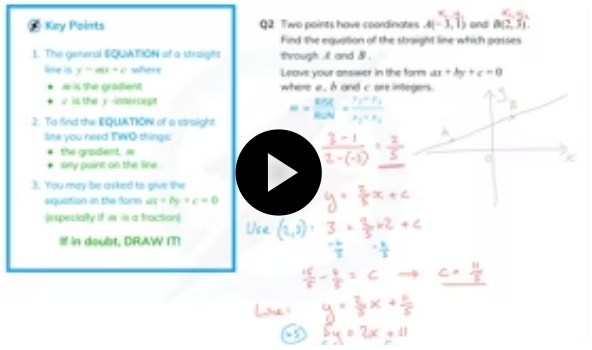

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Help me understand

Finding equations of straight lines

What is the equation of a straight line?

- The general **equation** of a **straight line** is $y = mx + c$ where
  - m is the **gradient**
  - c is the **y-intercept**
    - The value where it **cuts** the **y-axis**
- $y = 5x + 2$ is a straight line with
  - gradient 5
  - y-intercept 2
- $y = 3 - 4x$ is a straight line with
  - gradient -4
  - y-intercept 3

How do I find the equation of a straight line from a graph?

- Find the **gradient** by drawing a triangle and using
  - $\text{gradient} = \frac{\text{rise}}{\text{run}}$
    - **Positive** for uphill lines, **negative** for downhill

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **Substitute** these values into $y = mx + c$

What if no y-intercept is shown?

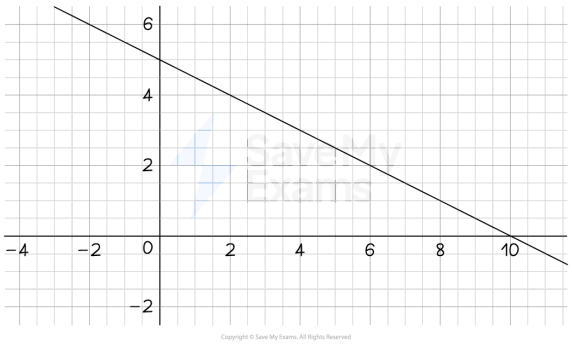
- If you **can't** read off the ***y*-intercept**
  - find any **point** on the line
  - **substitute** it into the **equation**
  - **solve** to **find c**
- For example, a line with gradient 6 passes through (2, 15)
  - The y-intercept is unknown
    - Write $y = 6x + c$
  - Substitute in $x = 2$ and $y = 15$
    - $15 = 6 \times 2 + c$
    - $15 = 12 + c$
  - Solve for c
    - $c = 3$
  - The equation is $y = 6x + 3$

What are the equations of horizontal and vertical lines?

- A **horizontal** line has the equation $y = c$
  - c is the y-intercept
- A **vertical** line has the equation $x = k$
  - k is the x-intercept
- For example
  - $y = 4$
  - $x = -2$

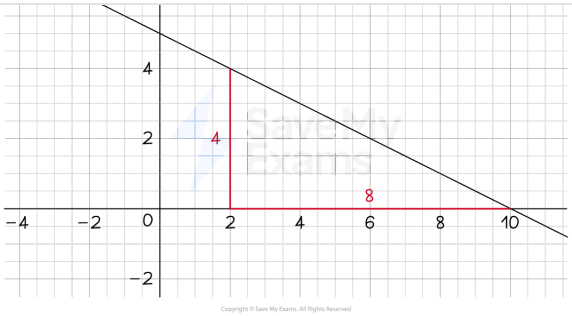
Worked Example

(a) Find the equation of the straight line shown in the diagram below.



Answer:

Find m, the gradient
Identify any two points the line passes through and work out the rise and run



The rise is 4
The run is 8

Calculate the fraction $\frac{\text{rise}}{\text{run}}$

$$\frac{\text{rise}}{\text{run}} = \frac{4}{8} = \frac{1}{2}$$

The slope is downward (downhill), so it is a negative gradient

$$\text{gradient, } m = -\frac{1}{2}$$

Now find the y-intercept
The line cuts the y-axis at 5

$$y\text{-intercept, } c = 5$$

Substitute the gradient, m, and the y-intercept, c, into $y = mx + c$

$$y = -\frac{1}{2}x + 5$$

(b) Find the equation of the straight line with a gradient of 3 that passes through the point (5, 4).

Answer:

Substitute $m = 3$ into $y = mx + c$
Leave c as an unknown letter

$$y = 3x + c$$

Substitute $x = 5$ and $y = 4$ into the equation
Solve the equation to find c

$$4 = 3 \times 5 + c$$
$$4 = 15 + c$$
$$-11 = c$$

You now know c
Replace c with −11 to complete the equation of the line

$$y = 3x - 11$$

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